Numerical Methods Group Project Instructions

CBE- GY 6153: Numerical Methods in Chemical Engineering

Submission due: Tuesday, May 5th, 2026 at 5:00 pm EST

This group project is designed to give you hands-on experience applying numerical methods to model a chemical engineering system using computational tools such as MATLAB or COMSOL. You will work in teams of 4 students to define your own modeling problem of interest, go through a structured development process, and present your work in both a written report and a final presentation.

Stage 1. Ideation and Instructor Consultation (20 points)

Select a chemical engineering system that you are interested in modeling. This can be something relevant to your research, a real-world process, a chemical device, or a theoretical system that can be represented using numerical methods.

Each team will schedule a meeting with the instructor to discuss their proposed system and modeling approach. To prepare for this consultation, teams must create a short slide deck with the following:

Slide 1: Description of the problem or system to be modeled. What is it, and why is it interesting or important?

Slide 2: Geometry and general setup of the problem (e.g., governing equations, boundary conditions, etc). Include a sketch or diagram.

Slide 3: Description of the computational methods the team plans to use (e.g., finite difference, finite element, numerical integration, etc.). Any initial thoughts on boundary conditions, input parameters, or simplifications that will be used in the model.

Each team will have 15 minutes to discuss their concept with the instructor and receive feedback.

Stage 2. Project Report (50 points)

Each team will submit a final written report presenting their modeling project. The report should be clear, concise, and structured like a short scientific paper. It should demonstrate the team's ability to apply numerical methods to a chemical engineering system using MATLAB or COMSOL.

The report must include the following sections:

<u>Title:</u> A descriptive title that reflects the system or problem modeled.

<u>Abstract (max 150 words):</u> A brief summary of the motivation, methods, and key results.

<u>Introduction and Motivation:</u> Describe the system of interest, why it is important or interesting, and what you aim to learn or demonstrate through modeling.

Methodology: Present your modeling approach. Include a description of the numerical methods used, any relevant equations, assumptions made, geometry, boundary conditions, and software tools.

Results and Discussion: Show the output of your model and discuss what the results mean. Include figures or plots as needed to support your discussion. Reflect on the accuracy, challenges, or limitations of your approach.

Conclusion: Summarize the main findings of the project and any broader insights.

References: Cite any sources or resources used.

Formatting Guidelines: The report must not exceed 10 pages, including figures and tables. Use single spacing and a minimum 11-point font (e.g., Times New Roman or Arial). Margins must be at least one inch on all sides. The References section does not count toward the page limit.

Final reports are due on May 5th via Brightspace. Only one submission per team is required for each stage. Be sure to include the names of all team members on every submission.

Stage 3. Presentation (30 points)

Each team will give a short in-class presentation summarizing their project. The presentation should be clear, focused, and mirror the structure of the written report.

Presentations should cover the following four elements:

1. Motivation – Briefly explain the problem or system and why it matters.

2. Methodology – Describe the geometry and modeling approach, including the numerical methods used.

3. Results – Present and explain the key modeling results, using figures or plots.

4. Conclusions – Summarize the main findings and takeaways from the project.

Presentations must not exceed 10 minutes, followed by 3–5 minutes of Q&A. All team members should contribute to preparing the presentation, though not everyone is required to speak. Presentations will take place in class on May 5th.